DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2026–27

ODD SEMESTER

INFORMATION ASSURANCE AND SECURITY

(24CS3105)

ALM – PROTOTYPE REVIEW

submitted by

P. Vyshnavy
2420090021

Cluster: 4 Section: 11

Course Instructor

Dr. Jagan Mohan Reddy Danda
Associate Professor



KONERU LAKSHMAIAH EDUCATION FOUNDATION, BOWRAMPET

Hyderabad–500043, Telangana, India

2026–27

Abstract

Information Assurance and Security focuses on protecting information and communication systems against unauthorized access, modification, and disclosure. This prototype review presents the implementation and study of different cryptographic algorithms including Caesar Cipher, Playfair Cipher, Hill Cipher, Vigenère Cipher, Simplified Data Encryption Standard (SDES), Advanced Encryption Standard (AES), Stream Cipher (RC4), Diffie–Hellman, and RSA. For each algorithm, the report provides its definition, working principle, mathematical notation, algorithmic steps, and prototype screenshot. The objective of this prototype is to understand the working of different encryption, decryption, and key-exchange techniques and demonstrate their practical implementation.

Contents

Abstract **1**

1 Introduction **6**
1.1 Objectives 6

2 Caesar Cipher **6**
2.1 Definition 6
2.2 Mathematical Notation 6
2.3 Algorithm 7
2.4 Prototype Screenshot 7

3 Playfair Cipher **7**
3.1 Definition 7
3.2 Working Principle 8
3.3 Algorithm 8
3.4 Prototype Screenshot 8

4 Hill Cipher **9**
4.1 Definition 9
4.2 Mathematical Notation 9
4.3 Algorithm 9
4.4 Prototype Screenshot 10

5 Vigenère Cipher **10**
5.1 Definition 10
5.2 Mathematical Notation 10
5.3 Algorithm 11
5.4 Prototype Screenshot 11

6 Simplified Data Encryption Standard (SDES) **11**
6.1 Definition 11
6.2 Mathematical Representation 11
6.3 Algorithm 12
6.4 Prototype Screenshot 12

7 Advanced Encryption Standard (AES) **13**
7.1 Definition 13
7.2 Key Sizes 13
7.3 Mathematical Representation 13
7.4 Algorithm 13

7.5 Prototype Screenshot 14

8 Stream Cipher (RC4) 14
8.1 Definition 14
8.2 Mathematical Notation 14
8.3 Algorithm 15
8.4 Prototype Screenshot 15

9 Diffie–Hellman Key Exchange 15
9.1 Definition 15
9.2 Mathematical Notation 15
9.3 Algorithm 16
9.4 Prototype Screenshot 17

10 RSA 17
10.1 Definition 17
10.2 Key Generation 17
10.3 Encryption 18
10.4 Decryption 18
10.5 Algorithm 18
10.6 Prototype Screenshot 19

11 Comparison of Cryptographic Algorithms 19

12 Prototype Implementation 19
12.1 Prototype Features 20

13 GitHub URL 20

List of Tables

1 AES Key Sizes 13
2 Comparison of Cryptographic Algorithms 19

List of Figures

1 Caesar Cipher Prototype 7
2 Playfair Cipher Prototype 8
3 Hill Cipher Prototype 10
4 Vigenère Cipher Prototype 11
5 SDES Prototype 12
6 AES Prototype 14
7 RC4 Prototype 15
8 Diffie–Hellman Prototype 17
9 RSA Prototype 19

1 Introduction

Cryptography is an important area of Information Assurance and Security. It provides techniques for protecting information by transforming readable data into an unreadable form using encryption algorithms and keys.

This prototype demonstrates different cryptographic algorithms that are studied in information security. The algorithms covered in this report include classical ciphers, symmetric-key encryption algorithms, stream ciphers, public-key cryptography, and key-exchange techniques.

1.1 Objectives

The objectives of this prototype are:

- To understand the basic concepts of cryptography.
- To implement different encryption and decryption algorithms.
- To study mathematical operations used in cryptography.
- To understand symmetric and asymmetric cryptography.
- To demonstrate encryption and decryption using a prototype.
- To understand secure key exchange using Diffie–Hellman.
- To understand RSA public-key encryption and decryption.

2 Caesar Cipher

2.1 Definition

The Caesar Cipher is a classical substitution cipher in which every letter in the plaintext is shifted by a fixed number of positions in the alphabet.

2.2 Mathematical Notation

The encryption formula is:

$$C = (P + K) \bmod 26 \tag{1}$$

The decryption formula is:

$$P = (C - K) \bmod 26 \tag{2}$$

where P is plaintext, C is ciphertext, and K is the key.

2.3 Algorithm

Algorithm 1 Caesar Cipher Encryption

1: Read plaintext P
2: Read key K
3: **for** each character in P **do**
4: Convert character into numerical value
5: $C \leftarrow (P + K) \bmod 26$
6: Convert C back into character
7: **end for**
8: Display ciphertext

2.4 Prototype Screenshot

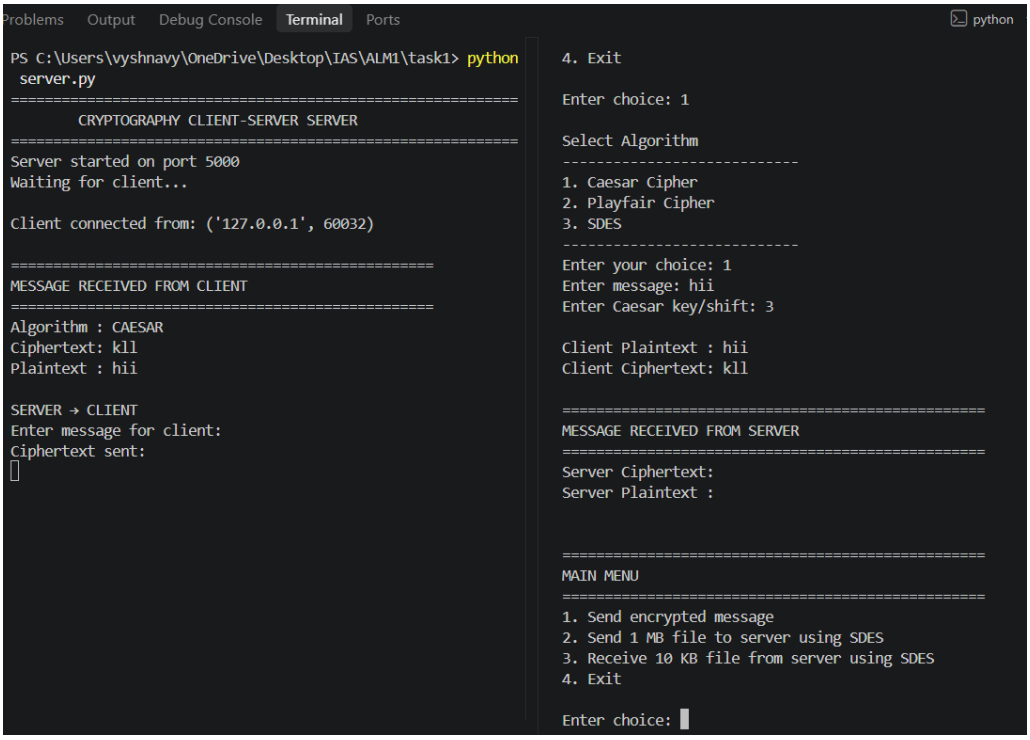


Figure 1: Caesar Cipher Prototype

3 Playfair Cipher

3.1 Definition

The Playfair Cipher is a digraph substitution cipher that encrypts two letters at a time using a 5×5 matrix generated from a keyword.

3.2 Working Principle

A Playfair matrix is created using a keyword and the remaining letters of the alphabet. The letters I and J are generally combined into one cell.

3.3 Algorithm

Algorithm 2 Playfair Cipher

```
Read plaintext and keyword
Generate 5 × 5 Playfair matrix
Divide plaintext into pairs
for each pair do
    if both letters are in the same row then
        Replace each letter with the letter to its right
    else if both letters are in the same column then
        Replace each letter with the letter below it
    else
        Form a rectangle
        Replace each letter with the letter at the other corner
    end if
end for
Display ciphertext
```

3.4 Prototype Screenshot

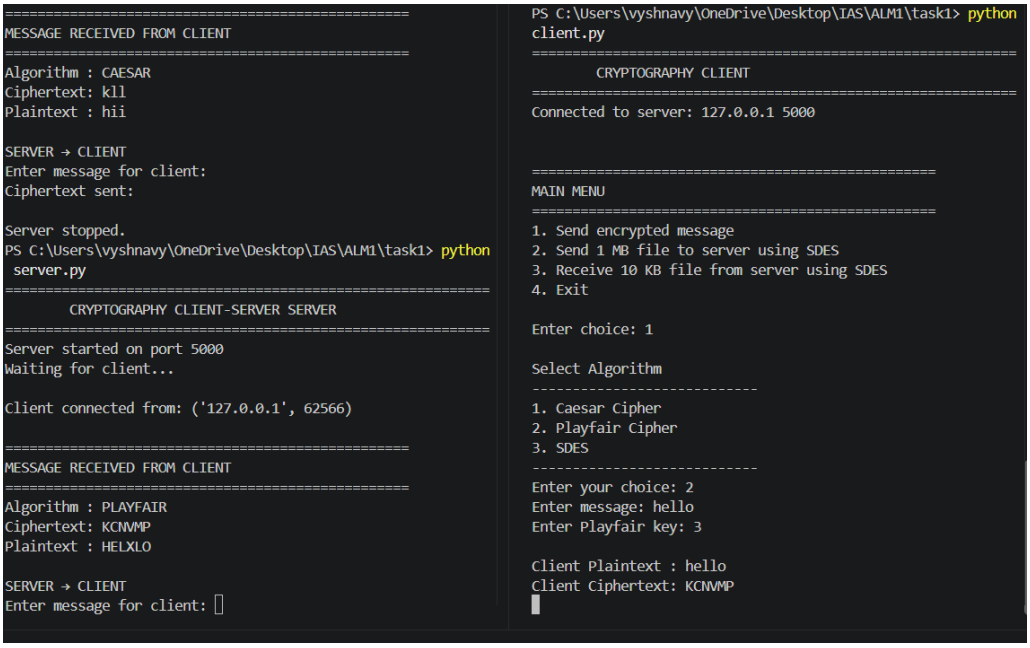


Figure 2: Playfair Cipher Prototype

4 Hill Cipher

4.1 Definition

The Hill Cipher is a polygraphic substitution cipher based on matrix multiplication and modular arithmetic.

4.2 Mathematical Notation

Encryption is performed using:

$$C = KP \bmod 26 \tag{3}$$

Decryption is:

$$P = K^{-1}C \bmod 26 \tag{4}$$

where K is the key matrix and K^{-1} is its modular inverse.

4.3 Algorithm

Algorithm 3 Hill Cipher

```
Read plaintext
Read key matrix K
Convert plaintext characters into numerical values
Divide plaintext into blocks
for each plaintext block P do
    Calculate C = KP mod 26
end for
Convert numerical values into characters
Display ciphertext
```

4.4 Prototype Screenshot

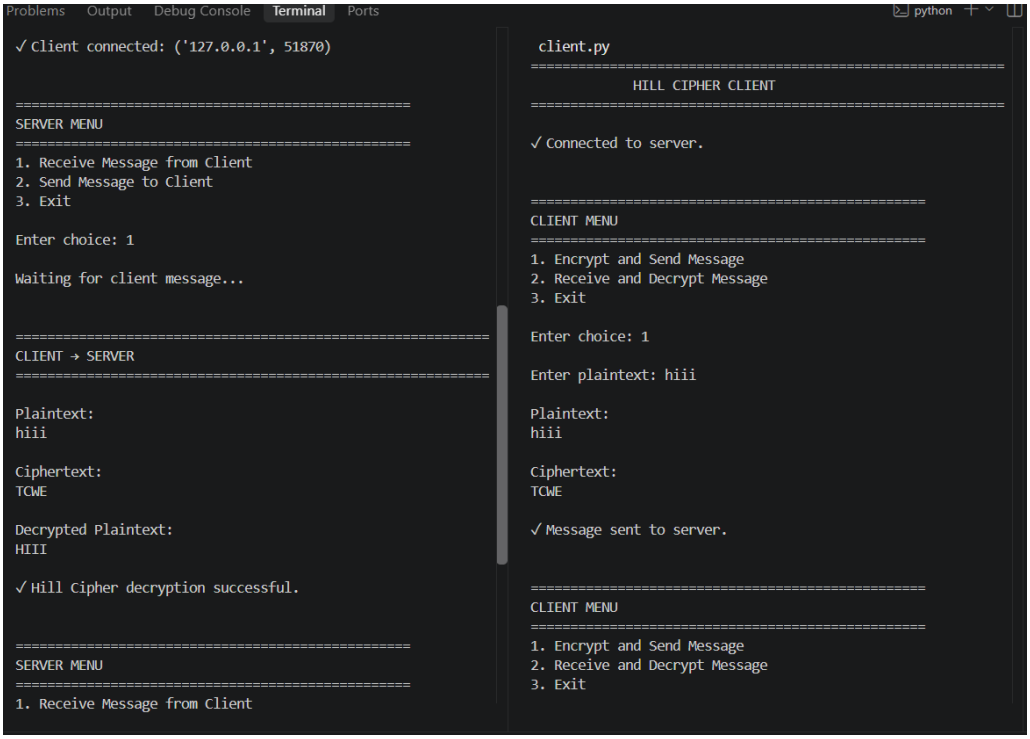


Figure 3: Hill Cipher Prototype

5 Vigenère Cipher

5.1 Definition

The Vigenère Cipher is a polyalphabetic substitution cipher that uses a keyword to perform multiple Caesar Cipher shifts.

5.2 Mathematical Notation

Encryption:

$$C_i = (P_i + K_i) \bmod 26 \tag{5}$$

Decryption:

$$P_i = (C_i - K_i) \bmod 26 \tag{6}$$

5.3 Algorithm

Algorithm 4 Vigenère Cipher

1: Read plaintext
2: Read keyword
3: Repeat keyword to match plaintext length
4: **for** each character **do**
5: Convert plaintext and key into numerical values
6: $C_i \leftarrow (P_i + K_i) \bmod 26$
7: **end for**
8: Display ciphertext

5.4 Prototype Screenshot

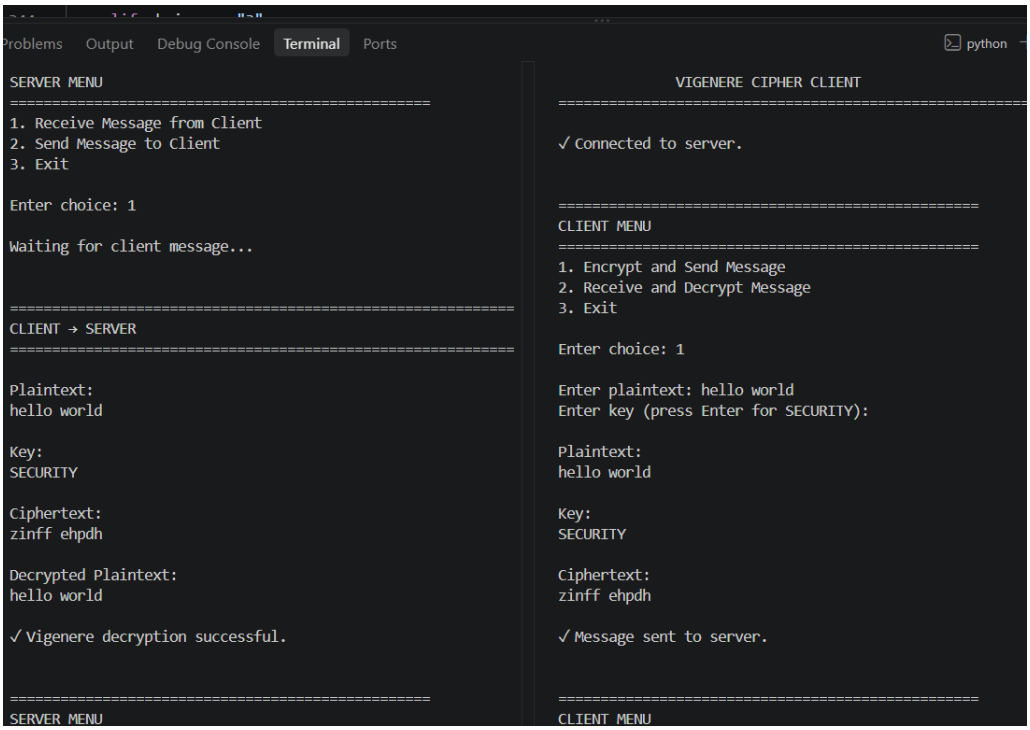


Figure 4: Vigenère Cipher Prototype

6 Simplified Data Encryption Standard (SDES)

6.1 Definition

Simplified Data Encryption Standard (SDES) is an educational block cipher designed to demonstrate basic block cipher operations.

6.2 Mathematical Representation

Encryption:

$$C = E_K(P)$$

Decryption:

$$P = D_K(C)$$

where P is plaintext, C is ciphertext, and K is the key.

6.3 Algorithm

Algorithm 5 SDES Encryption

1: Generate subkeys from the original key
2: Apply initial permutation
3: Perform first encryption round
4: Swap left and right halves
5: Perform second encryption round
6: Apply inverse permutation
7: Output ciphertext

6.4 Prototype Screenshot

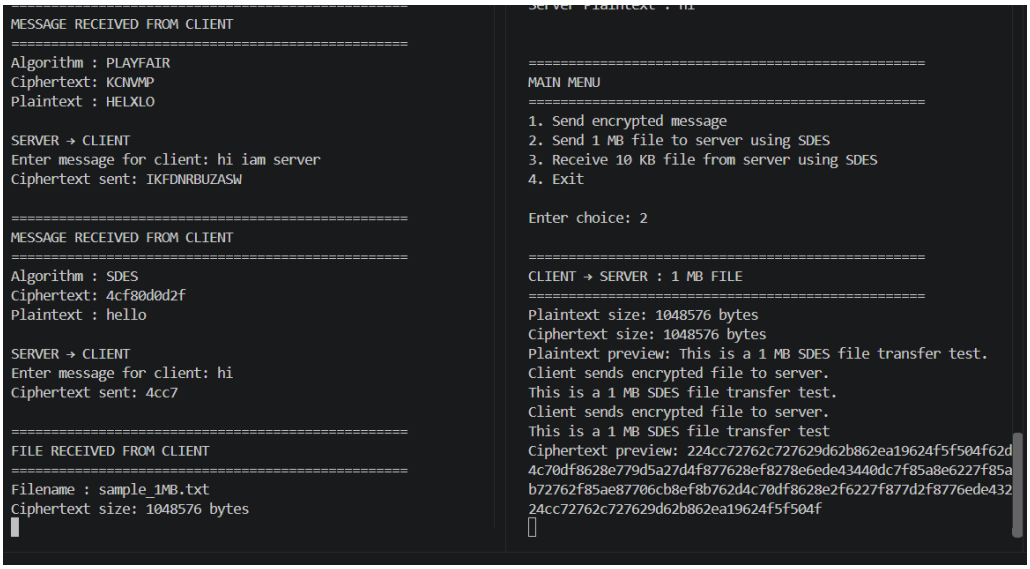


Figure 5: SDES Prototype

7 Advanced Encryption Standard (AES)

7.1 Definition

Advanced Encryption Standard (AES) is a symmetric-key block cipher used to protect digital information.

7.2 Key Sizes

Table 1: AES Key Sizes

AES Version	**Key Size**
AES-128	128 bits
AES-192	192 bits
AES-256	256 bits

7.3 Mathematical Representation

$$C = E_K(P)$$

$$P = D_K(C)$$

7.4 Algorithm

Algorithm 6 AES Encryption

```
Expand the encryption key
Perform AddRoundKey
for each encryption round do
    Perform SubBytes
    Perform ShiftRows
    Perform MixColumns
    Perform AddRoundKey
end for
Perform final round without MixColumns
Output ciphertext
```

7.5 Prototype Screenshot

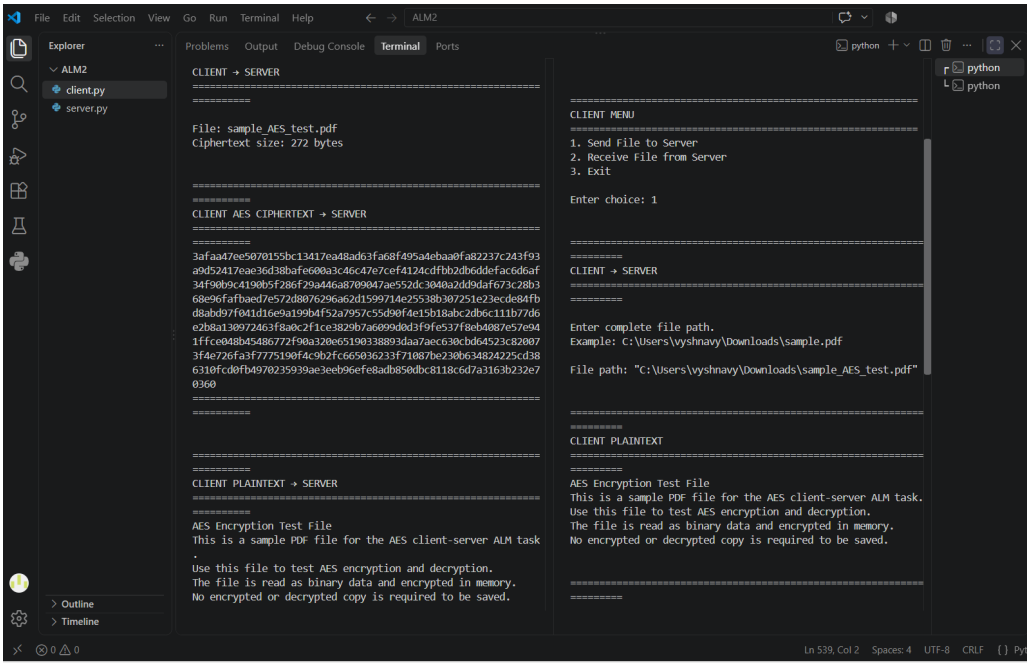


Figure 6: AES Prototype

8 Stream Cipher (RC4)

8.1 Definition

RC4 is a stream cipher that generates a pseudorandom keystream and combines it with plaintext using the XOR operation.

8.2 Mathematical Notation

Encryption:

$$C_i = P_i \oplus K_i \tag{7}$$

Decryption:

$$P_i = C_i \oplus K_i \tag{8}$$

8.3 Algorithm

Algorithm 7 RC4 Stream Cipher

1: Initialize the state array
2: Perform Key Scheduling Algorithm (KSA)
3: Generate the keystream
4: XOR plaintext with the keystream
5: Output ciphertext

8.4 Prototype Screenshot

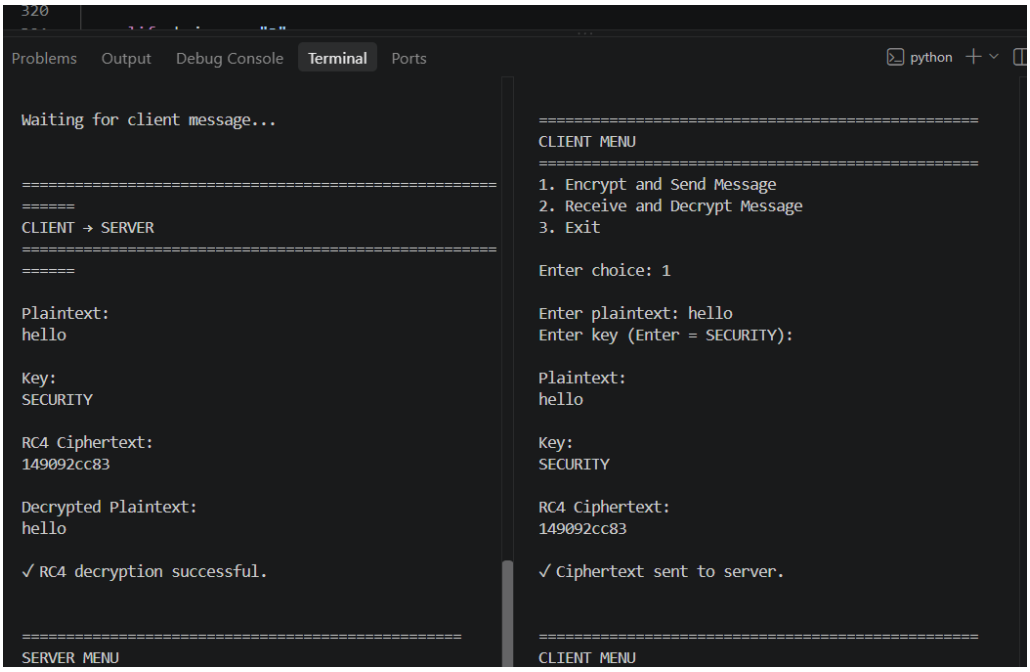


Figure 7: RC4 Prototype

9 Diffie–Hellman Key Exchange

9.1 Definition

Diffie–Hellman is a key exchange method that allows two parties to establish a shared secret key over an insecure communication channel.

9.2 Mathematical Notation

Alice selects a private key a:

$$A = g^a \bmod p \tag{9}$$

Bob selects a private key b:

$$B = g^b \bmod p \tag{10}$$

Alice calculates:

$$S = B^a \bmod p \tag{11}$$

Bob calculates:

$$S = A^b \bmod p \tag{12}$$

Therefore:

$$B^a \bmod p = A^b \bmod p$$

9.3 Algorithm

Algorithm 8 Diffie–Hellman Key Exchange

1: Select public prime number p
2: Select public generator g
3: Alice selects private key a
4: Bob selects private key b
5: Calculate $A = g^a \bmod p$
6: Calculate $B = g^b \bmod p$
7: Alice calculates $S = B^a \bmod p$
8: Bob calculates $S = A^b \bmod p$
9: Verify that both shared secrets are equal

9.4 Prototype Screenshot

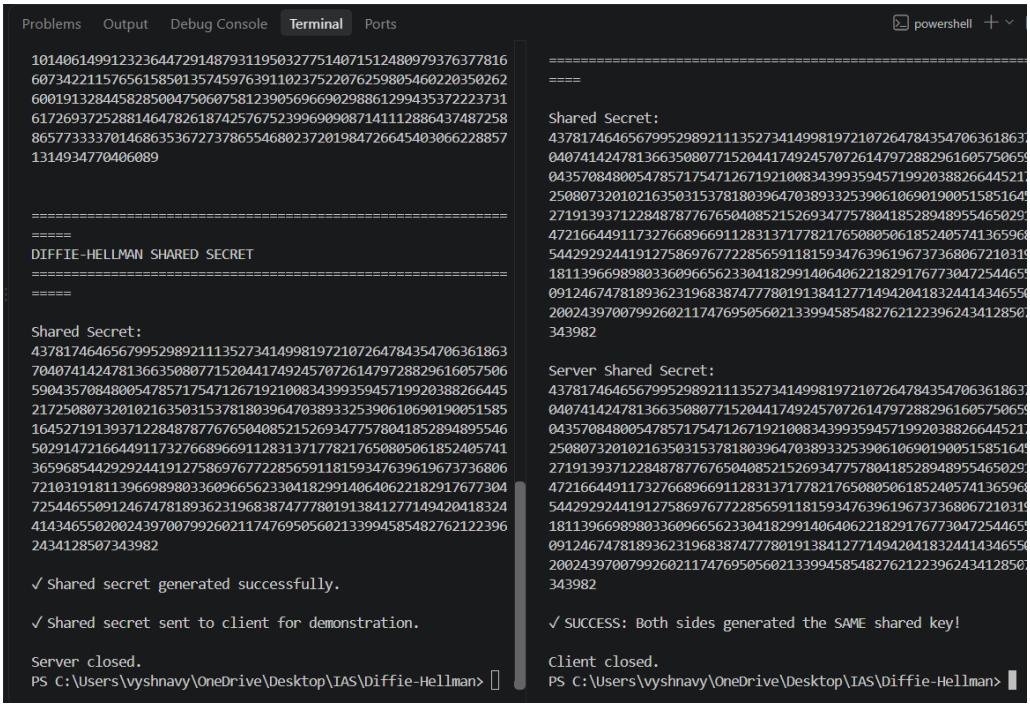


Figure 8: Diffie–Hellman Prototype

10 RSA

10.1 Definition

RSA is a public-key cryptographic algorithm that uses a pair of keys: a public key for encryption and a private key for decryption.

10.2 Key Generation

Select two prime numbers p and q.

$$n = pq \tag{13}$$

$$\phi(n) = (p-1)(q-1) \tag{14}$$

Choose e such that:

$$gcd(e, \phi(n)) = 1$$

Calculate d:

$$ed \equiv 1 \pmod{\phi(n)} \tag{15}$$

The public key is:

$$(e, n)$$

The private key is:

$$(d, n)$$

10.3 Encryption

$$C = M^e \bmod n \tag{16}$$

10.4 Decryption

$$M = C^d \bmod n \tag{17}$$

10.5 Algorithm

Algorithm 9 RSA Algorithm

1: Select two prime numbers p and q
2: Calculate $n = pq$
3: Calculate $\phi(n) = (p-1)(q-1)$
4: Select e such that $gcd(e, \phi(n)) = 1$
5: Calculate d such that $ed \equiv 1 \pmod{\phi(n)}$
6: Generate public key (e, n)
7: Generate private key (d, n)
8: Encrypt using $C = M^e \bmod n$
9: Decrypt using $M = C^d \bmod n$
10: Display plaintext

10.6 Prototype Screenshot

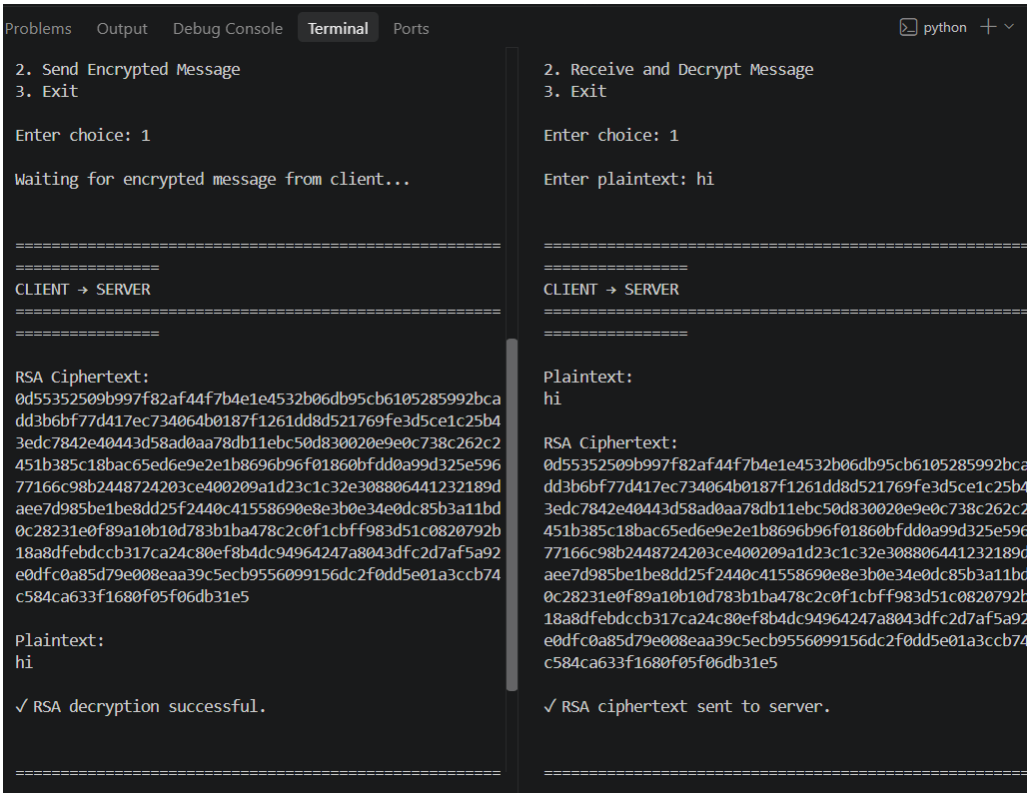


Figure 9: RSA Prototype

11 Comparison of Cryptographic Algorithms

Table 2: Comparison of Cryptographic Algorithms

Algorithm	**Type**	**Purpose**	**Key Concept**
Caesar Cipher	Classical	Encryption	Character shifting
Playfair Cipher	Classical	Encryption	Digraph substitution
Hill Cipher	Classical	Encryption	Matrix multiplication
Vigenère Cipher	Classical	Encryption	Polyalphabetic substitution
SDES	Symmetric	Encryption	Block cipher operations
AES	Symmetric	Encryption	Block cipher
RC4	Symmetric	Encryption	Stream cipher
Diffie–Hellman	Key Exchange	Key establishment	Shared secret
RSA	Asymmetric	Encryption	Public/private keys

12 Prototype Implementation

The prototype integrates multiple cryptographic algorithms into a single implementation. Users can select an algorithm, provide the required input and key values, and perform encryption or decryption.

12.1 Prototype Features

- Algorithm selection.
- Encryption and decryption operations.
- Key input and processing.
- Mathematical calculations for classical ciphers.
- Symmetric encryption using AES and RC4.
- Key exchange using Diffie–Hellman.
- Public-key encryption using RSA.
- Output display for verification.

13 GitHub URL

The complete source code of the prototype is available at:

https://github.com/vyshnavy2025/IAS_2420090021